

BIOGRAPHICAL SKETCH

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NAME MacIver, M., Bruce	POSITION TITLE Associate Professor of Neurophysiology		
eRA COMMONS USER NAME MACIVER.BRUC			
EDUCATION/TRAINING (Begin with baccalaureate or other initial professional education, such as nursing, and include postdoctoral training.)			
INSTITUTION AND LOCATION	DEGREE (if applicable)	YEAR(s)	FIELD OF STUDY
University of Calgary, Alberta, Canada	BSc, MSc	1981	Pharmacology
University of Calgary, Alberta, Canada	PhD	1987	Neuroscience
Stanford University, Palo Alto, California	PostDoc	1987 - 1989	Neuropharmacology
Stanford University, Palo Alto, California	PostDoc	1989 - 1991	Neurophysiology

Professional Positions:

December 1991 to June 1998 Assistant Professor of Neurophysiology, Department of Anesthesia, Stanford University School of Medicine.

June 1998 to present Associate Professor of Neurophysiology, Department of Anesthesia, Stanford University School of Medicine.

Affiliated: Bioengineering/Biodesign and the Neuroscience Institute

Awards and Other Professional Activities:

1983 to 1986 Alberta Heritage Foundation for Medical Research: Studentship

1983 Mental Health Advisory Council of Alberta: Research Scholarship

1988 Stanford University School of Medicine: Research Honors Award

1988 Pharmacological Society of Canada Award for Outstanding Research

2000 Elected to Association of University Anesthesiologists

2004 Appointed to the NIH, Adjunct Reviewer

2004 Allen V. Cox Medal for Undergraduate Research Training, Stanford University

2005 Anesthesia and Analgesia top 100 citations

2008 Appointed to the NIH NIDA summer minority student training program.

Relevant Publications: (Total of 51 papers/chapters, 105 abstracts; since 1982)

1. MacIver, M.B., Roth, S.H. (1987) Enflurane-induced burst firing of hippocampal CA1 neurons *in vitro*. Br. J. Anaesth 59:369-78.
2. MacIver, M.B., Roth, S.H. (1987) Anaesthetics produce differential actions on the discharge activity of a single neuron. Eur. J. Pharmacol. 139:43-52.
3. Konopacki, J., Bland, B.H., MacIver, M.B., Roth, S.H. (1987) Cholinergic theta rhythm in transected hippocampal slices: independent CA1 and dentate generators. Brain Res 436:217-22.
4. MacIver, M.B., Roth, S.H. (1988) Inhalation anaesthetics exhibit pathway-specific and differential actions on hippocampal synaptic responses *in vitro*. Br. J. Anaesth, 60:680-91.
5. Kendig, J.J., Grossman, Y., MacIver, M.B. (1988) Pressure reversal of anaesthesia: a synaptic mechanism. Br. J. Anaesth, 60:806-16.
6. MacIver, M.B., Tauck, D.L., Kendig, J.J. (1988) General anaesthetic modification of synaptic facilitation and long-term potentiation in hippocampus. Br. J. Anaesth. 61:301-10.
7. Mody, I., MacIver, M.B., Tanelian, D.L. (1991) Halothane enhances tonic neuronal inhibition by elevating intracellular calcium. Brain Res, 538:319-23.
8. Kendig, J.J., MacIver, M.B., Roth, S.H. (1991) Anesthetic actions in the hippocampal formation. Ann. N. Y. Acad. Sci., 625:37-53.

9. Tanelian, D.L., Kosek, P., Mody, I., MacIver, M.B. (1993) The role of the GABA receptor/chloride channel complex in anesthesia. *Anesthesiology*, 78(4):757-76.
10. MacIver, M.B., Tanelian, D.L. (1993) Free nerve ending terminal morphology is fiber type specific for A δ and C fibers innervating rabbit corneal epithelium. *J Neurophysiol*, 69(5):1779-83.
11. MacIver, M.B., Tanelian, D.L. (1993) Structural and functional specialization of A δ and C fiber free nerve endings innervating rabbit corneal epithelium. *J Neurosci*, 13(10):4511-24.
12. MacIver, M.B. (1997) General Anesthetic Actions at Glutamate and GABA Synapses. In Anesthesia: Biological Foundations, edited by Drs: J.F. Biebuyck, C. Lynch III, M. Maze, L.J. Saidman, T.L. Yaksh, and W.M. Zapol, Lippincott-Raven, Philadelphia, PA, pp: 277-286.
13. MacIver, M.B., Mandema, J.W., Stanski, D.R., Bland, B.H. (1996) Thiopental uncouples hippocampal and cortical synchronized EEG activity. *Anesthesiology*, 84(6):1411-24.
14. Lukatch, H.S., MacIver, M.B. (1996) Synaptic mechanisms of thiopental-induced alterations in synchronized cortical activity. *Anesthesiology*, 84(6):1425-34.
15. MacIver, M.B., Amagasu, S.M., Mikulec, A.A., Monroe F.A. (1996) Riluzole anesthesia: use-dependent block of presynaptic glutamate fibers. *Anesthesiology*, 85(3):626-34.
16. MacIver, M.B., Mikulec, A.A., Amagasu, S.M., Monroe, F.A. (1996) Volatile anesthetics depress glutamate transmission *via* presynaptic actions. *Anesthesiology*, 85(4):823-34.
17. Lukatch, H.S., MacIver, M.B. (1997) Physiology, pharmacology and topography of cholinergic neocortical oscillations *in vitro*. *J Neurophysiol*, 77:2427-2445.
18. Lukatch, H.S., MacIver, M.B. (1997) Voltage clamp analysis of halothane effects on GABA_AFAST and GABA_ASLOW inhibitory currents. *Brain Res*, 765(1):108-112.
19. Lukatch, H.S., Echon, R.M., MacIver, M.B., Werchan, P.M. (1997) G-force induced alterations in rat EEG activity: a quantitative analysis. *Electroencephalogr Clin Neurophysiol, The EEG Journal*, 103(5):563-573.
20. Hagan, C.E., Trudell, J.R., Pearce, R.A., MacIver, M.B. (1998) Concentration measures of volatile anesthetics in the aqueous phase using calcium sensitive electrodes. *J Neurosci Methods*, 81(1-2):177-184.
21. Mikulec, A.A., Pittson, S., Amagasu, S.M., Monroe, F.A., MacIver, M.B. (1998) Halothane depresses action potential conduction in hippocampal axons. *Brain Res*, 796(1-2):231-238.
22. Rabinovici, G.D., Lukatch, H.S., MacIver, M.B. (2000) Ischemic modulation of cortical micro-EEG activity in rat brain slices. *Clin Neurophysiol*, 111(1):112-121.
23. Nishikawa, K., MacIver, M.B. (2000) Excitatory synaptic transmission mediated by NMDA receptors is more sensitive to isoflurane than are non-NMDA receptor-mediated responses. *Anesthesiology*, 92(1):228-236.
24. Nishikawa, K., MacIver, M.B. (2000) Membrane and synaptic actions of halothane on rat hippocampal pyramidal neurons and inhibitory interneurons. *J Neurosci*, 20(16):5915-5923.
25. Lingameneni, R., Krasowski, M.D., Jenkins, A., Pashkov, V., Truong, T., MacIver, M.B., Harrison, N.L., Hemmings Jr., H.C. (2001) Anesthetic properties of 4-iodopropofol: Implications for theories of mechanisms of anesthesia. *Anesthesiology*, 94(6):1050-1057.
26. Nishikawa K, MacIver MB. (2001) Agent-selective effects of volatile anesthetics on GABA_A receptor-mediated synaptic inhibition in hippocampal interneurons. *Anesthesiology*, 94(2):340-347.
27. Bieda, M.C., MacIver, M.B. (2004) A Major Role For Tonic GABA_A Conductances In Anesthetic Suppression Of Intrinsic Neuronal Excitability, *J Neurophysiol*, 92:1658-1667.
28. Gredell, J.A., Turnquist, P.A., MacIver, M.B., Pearce R.A. (2004) Determination of diffusion and partition coefficients of propofol in rat brain tissue: implications for studies of drug action *in vitro*. *Br J Anaesth*. Dec; 93(6):810-7.
29. Pittson, S., Himmel, A.M., MacIver, M.B. (2004) Multiple synaptic and membrane sites of anesthetic action in the CA1 region of rat hippocampal slices. *BMC Neuroscience*, 5:52-59.
30. Lukatch, H.S., Kiddoo, C.E., MacIver, M.B. (2005) Anesthetic-induced burst suppression EEG activity requires glutamate mediated excitatory synaptic transmission. *Cereb Cortex*, 15:1322-1331.
31. Sceniak M.P., MacIver, M.B. (2006) Anesthesia *in silico*. *Anesthesiology* 104:400-2.
32. Winegar, B.D., MacIver, M.B. (2006) Isoflurane depresses hippocampal CA1 glutamate nerve terminals without inhibiting fiber volleys. *BMC Neuroscience* 7:5-12.
33. Sceniak M.P., MacIver, M.B. (2006) Cellular actions of urethane on rat visual cortical neurons *in vitro*. *J. Neurophysiology* 95:3865-74.
34. Sceniak M.P., MacIver, M.B. (2008) Slow GABA(A) mediated synaptic transmission in rat visual cortex. *BMC Neurosci*. 2008 Jan 16;9:8.
35. Bieda M., Su H., MacIver M.B. (2008) Anesthetics discriminate between tonic and phasic GABA(A) receptors. *Anesthesia and Analgesia* (in press).

Research Projects Ongoing or Completed During the Last 3 Years:

“Anesthetic Actions at GABA and Glutamate Synapses”

Principal Investigator: M. Bruce MacIver

Agency: NIH - NIGMS

Type: R01 (GM54767, Years 0-11) Period: May 1, 1996 to April 3, 2015
Pending Competing Renewal to Start in 2009

The major goals of this project are 1) To determine the effective concentration range for anesthetic-induced block of LTP. 2) To determine the involvement of GABA and glutamate-mediated synapses in anesthetic-induced block of LTP. 3) To determine whether postsynaptic actions at or downstream from NMDA receptors contribute to the anesthetic-induced block of LTP.

“Abused Inhalant Actions at GABA and Glutamate Synapses”

Principal Investigator: M. Bruce MacIver

Agency: NIH - NIDA

Type: R01 (DA017884-01, Years 0-4) Period: July 1, 2005 to August 31, 2009
Awarded

The major goals of this project are: 1) To determine whether inhaled solvents alter GABA transmission, 2) To determine whether solvents alter glutamate transmission, and 3) To determine the detailed mechanisms of action for solvent effects at amino acid synapses.